more familiar everyday strings. But since fundamental strings are so tiny, this approach cannot be carried out and a more indirect method is called for. In 1974, when Scherk and Schwarz proposed that one particular pattern of string vibration was the graviton particle, they were able to exploit such an indirect approach and thereby predict the tension on the strings of string theory. Their calculations revealed that the strength of the force transmitted by the proposed graviton pattern of string vibration is inversely proportional to the string's tension. And since the graviton is supposed to transmit the gravitational force—a force that is intrinsically quite feeble—they found that this implies a colossal tension of a thousand billion billion billion billion (1039) tons, the so-called Planck tension. Fundamental strings are therefore extremely stiff compared with more familiar examples. This has three important consequences.

Three Consequences of Stiff Strings

First, whereas the ends of a violin or a piano string are pinned down, ensuring that they have a fixed length, no analogous constraining frame pins down the size of a fundamental string. Instead, the huge string tension causes the loops of string theory to contract to a minuscule size. Detailed calculation reveals that being under Planck tension translates into a typical string having Planck length—10-33 centimeters—as previously mentioned.8

Second, because of the enormous tension, the typical energy of a vibrating loop in string theory is extremely high. To understand this, we note that the greater the tension a string is under, the harder it is to get it to vibrate. For instance, it's far easier to pluck a violin string and set it vibrating than it is to pluck a piano string. Two strings, therefore, that are under different tension and are vibrating in precisely the same way will not have the same energy. The string with higher tension will have more energy than the string with lower tension, since more energy must be exerted to set it in motion.

This alerts us to the fact that the energy of a vibrating string is determined by two things: the precise manner in which it vibrates (more frantic patterns corresponding to higher energies) and the tension of the string (higher tension corresponding to higher energy). At first, this description might lead you to think that by taking on ever gentler vibrational patterns—patterns with ever smaller amplitudes and